

LLMs for AI

In this assignment you will pose instances of AI problems to LLMs, problems such as those we have covered in class or you have solved in previous homework assignments.

For example, in class we posed the ‘rider-order allocation’ problem to two LLMs (chatGPT and Claude), with the prompt:

Consider the following road distances between locations: A-B: 2, A-C: 4, B-C:1, C-D:2, C-E: 3, D-F: 3, D-E: 2, E-F: 2 There are two riders R1 at A and R2 at D available to deliver three orders: O1 from B to E, O2 from C to F, and O3 from A to D. What is the optimal allocation of orders to riders in two cases (a) to minimise the total fuel spent for all orders and (b) to minimise the elapsed time for all orders to be delivered.

You need to pose multiple (different) *instances* of a particular problem to the LLM and then **analyse** its outputs: (i) how often does it solve the problem (ii) what kind of errors does it make and (iii) is one LLM better than the other. (iv) compare the solutions arrived at by the LLM with those computed using algorithms (where applicable, i.e. for problems 2. and 3. below)

You can use free versions of chatGPT, Gemini and or Claude as available via their respective web-based interfaces. (Those who have their own API keys are free to automate the process and call the same LLMs programmatically, but the prompts must be English descriptions of instances even if programmatically generated.)

Do the above for the following problems:

1. Rider-order allocation problem above, i.e., variations of the example given above.
2. Container loading problem as in HW1 (make sure to the overall problem as well as specific instances in English, i.e., **not** using program-like structures e.g. JSON etc.)
3. Kabaddi game as in HW3, but in this case you need to play with the LLM yourself. (Those of you choosing to use an API approach can have your HW2 solutions play against the LLM, but this is optional).

What to submit:

Submit a pdf slide deck and recording of your presentation of the same, that includes:

- (a) Transcripts of your (or your program’s) conversations with LLMs (note exact model)
- (b) Your analysis of the behaviour of the LLMs under different scenarios.
- (c) Your conclusions regarding LLMs reasoning abilities on AI problems, as of today.

*As earlier, submissions must be **recorded by you** (presentation using ICAPP is optional.)*